

ContractIQ: Contract Obligation Tracking & Compliance Management Platform

1. Database Design Objective

The objective is to design a relational database structure for the ContractIQ application that supports its core modules and future backend implementation.

The database design focuses on identifying the required tables, defining the important columns and their appropriate PostgreSQL data types, identifying primary keys and foreign keys, and establishing the relationships between the tables. The design must also represent the complete database structure through an Entity Relationship (ER) diagram.

2. Required Database Tables

The database design specifies nine tables for the ContractIQ application. These tables support user management, contract management, document versioning, obligation tracking, renewal management, notifications, reporting, audit logging, and activity tracking.

Table	Purpose
users	Stores user information required for authentication, profile management, and role-based access control.
contracts	Stores information about contracts managed by the ContractIQ system.
contract_versions	Stores different versions of contracts/documents and supports version management.
obligations	Stores contractual obligations and information required to track their due dates, responsibility, completion, and progress.
renewals	Stores information related to contract renewals, expiry monitoring, renewal history, and renewal status.
notifications	Stores notifications related to renewals, obligations, compliance, and contract approval events.

reports	Supports the storage/management of information related to contract, compliance, renewal, obligation, and audit reports.
audit_logs	Stores audit-related information such as user actions, contract history, approval logs, security logs, and changes.
activities	Stores application/user activity information used for activity tracking and activity displays.

3. Detailed Database Schema

3.1 users Table

The **users** table stores information about users who access the ContractIQ application. It supports the project's requirements for user registration, secure authentication, profile management, and role-based access control. The project defines six user roles: Administrator, Legal Manager, Compliance Officer, Contract Manager, Department Head, and Employee.

Column	Data Type	Constraint
id	INTEGER	Primary Key, NOT NULL
full_name	VARCHAR(100)	NOT NULL
email	VARCHAR(255)	NOT NULL
role	VARCHAR(50)	NOT NULL
is_active	BOOLEAN	NOT NULL
password	VARCHAR(255)	NOT NULL

3.2 contracts Table

The **contracts** table represents the contracts managed by the ContractIQ application. The project requires contract creation, editing, review, approval workflow, assignment, status management, storage, categorization, search, and archival functionality.

Column	Data Type	Constraint
id	INTEGER	Primary Key, NOT NULL
title	VARCHAR(255)	NOT NULL
contract_number	VARCHAR(100)	NOT NULL
category	VARCHAR(100)	NOT NULL
status	VARCHAR(50)	NOT NULL
start_date	DATE	-
end_date	DATE	-
created_by	INTEGER	Foreign Key [users(id)]
assigned_to	INTEGER	Foreign Key [users(id)]

3.3 contract_versions Table

The **contract_versions** table stores different versions of contracts in the ContractIQ system. It supports document version control by maintaining multiple versions of the same contract and preserving their version history.

Column	Data Type	Constraint
id	INTEGER	PK, NOT NULL
contract_id	INTEGER	FK [contracts(id)], NOT NULL
version_number	INTEGER	NOT NULL
document_path	VARCHAR(500)	NOT NULL
created_by	INTEGER	FK [users(id)],NOT NULL
created_at	TIMESTAMP	NOT NULL

3.4 obligations Table

The **obligations** table stores the contractual obligations that need to be tracked and monitored in the ContractIQ system. It supports obligation registration, due-date tracking, responsibility assignment, completion status, compliance monitoring, and progress tracking.

Column	Data Type	Constraint
id	INTEGER	PK, NOT NULL
contract_id	INTEGER	FK [contracts(id)], NOT NULL
title	VARCHAR(255)	NOT NULL
description	TEXT	-
obligation_type	VARCHAR(100)	NOT NULL
due_date	DATE	NOT NULL
assigned_to	INTEGER	FK [users(id)]
status	VARCHAR(50)	NOT NULL
progress	INTEGER	-

3.5 renewals Table

The **renewals** table stores information related to the renewal of contracts in the ContractIQ system. It supports renewal tracking, expiry monitoring, renewal approval, renewal history, and renewal status management.

Column	Data Type	Constraint
id	INTEGER	PK, NOT NULL
contract_id	INTEGER	FK [contracts(id)], NOT NULL
renewal_date	DATE	NOT NULL
status	VARCHAR(50)	NOT NULL
approval_status	VARCHAR(50)	-
notes	TEXT	-

3.6 notifications Table

The **notifications** table stores notifications generated by the ContractIQ system for important contract-related events. It supports renewal reminders, obligation due alerts, compliance notifications, and contract approval alerts through email, SMS, and in-app notifications.

Column	Data Type	Constraint
id	INTEGER	PK, NOT NULL
user_id	INTEGER	FK [users(id)], NOT NULL
notification_type	VARCHAR(50)	NOT NULL
message	TEXT	NOT NULL
channel	VARCHAR(30)	NOT NULL
is_read	BOOLEAN	NOT NULL
created_at	TIMESTAMP	NOT NULL

3.7 reports Table

The **reports** table stores information about reports generated by the ContractIQ system. It supports different report types, including contract, compliance, renewal, obligation, and audit reports, and helps manage generated report information.

Column	Data Type	Constraint
id	INTEGER	PK, NOT NULL
report_type	VARCHAR(50)	NOT NULL
generated_by	INTEGER	FK [users(id)] ,NOT NULL
file_format	VARCHAR(20)	NOT NULL
file_path	VARCHAR(500)	-
created_at	TIMESTAMP	NOT NULL

3.8 audit_logs Table

The **audit_logs** table stores records of important activities and changes that need to be retained for auditing and security purposes. It supports user activity logs, contract history, approval logs, security logs, and change history required by the ContractIQ project

Column	Data Type	Constraint
id	INTEGER	PK, NOT NULL
user_id	INTEGER	FK [users(id)], NOT NULL
action	VARCHAR(100)	NOT NULL
entity_type	VARCHAR(50)	NOT NULL
details	TEXT	-
created_at	TIMESTAMP	NOT NULL

3.9 activities Table

The **activities** table records general user and system activities performed within the ContractIQ application. It helps maintain an activity history that can be displayed through dashboards and used for monitoring application usage.

Column	Data Type	Constraint
id	INTEGER	PK, NOT NULL
user_id	INTEGER	FK [users(id)], NOT NULL
activity_type	VARCHAR(50)	NOT NULL
description	TEXT	NOT NULL
created_at	TIMESTAMP	NOT NULL

4. Primary Keys and Foreign Keys

Table	Primary Key (PK)	Foreign Key (FK)	References
users	id	-	-
contracts	id	created_by	users(id)
contracts	id	assigned_to	users(id)
contract_versions	id	contract_id	contracts(id)
contract_versions	id	created_by	users(id)
obligations	id	contract_id	contracts(id)
obligations	id	assigned_to	users(id)
renewals	id	contract_id	contracts(id)
notifications	id	user_id	users(id)
reports	id	generated_by	users(id)
audit_logs	id	user_id	users(id)
activities	id	user_id	users(id)

5. Table Relationships

The ContractIQ database primarily uses **one-to-many (1:N) relationships**. Foreign keys connect related records and maintain consistency between the tables.

Parent Table	Child Table	Foreign Key	Relationship
users	contracts	created_by , assigned_to	1:N
contracts	contract_versions	contract_id	1:N
users	contract_versions	created_by	1:N
contracts	obligations	contract_id	1:N
users	obligations	assigned_to	1:N

contracts	renewals	contract_id	1:N
users	notifications	user_id	1:N
users	reports	generated_by	1:N
users	audit_logs	user_id	1:N
users	activities	user_id	1:N

6. ER Diagram

